

analysis

Cross-Artifact Analysis — Workshop Curriculum Platform

Feature: specs/001-workshop-curriculum-platform | **Date:** 2026-09-01
Scope: spec, plan, research (×4), data-model, contracts (×3), quickstart, tasks, checklist, project constitution — 8,433 lines.

Severity counts

Severity	Count
CRITICAL	4
HIGH	17
MEDIUM	22
LOW	6
Total	49

Coverage headline

- **FRs with no covering task (2):** FR-006, FR-039
- **FRs partially covered (7):** FR-005, FR-011, FR-012, FR-016, FR-017, FR-018, FR-029
- **SCs with no executing task (3):** SC-007, SC-008, SC-016a
- **SC partially validated (1):** SC-018
- **Scope creep: 0.** Four tasks trace to a decision/anchor rather than an FR (T003, T026, T039, T074) — all justified.
- **tasks.md's own count is correct by enumeration:**
 $7+9+16+10+14+14+6+15 = 91$.

Tree facts measured during analysis (not read off a document):

.gitmodules declares 9 submodules including a populated submodules/containers; workshop/ carries all four carriers; workshop/chapters/01/ holds 36 parts; the reference control plane is ai_interviewing/platform/scripts/, not ai_interviewing/scripts/; workshop/docs/ already exists.

CRITICAL

C1 — Container orchestration: two incompatible architectures on a claim the tree disproves

plan.md:73 and tasks.md:123,145 assume submodules/containers exists and is consumed. But research.md:281, contracts/pipeline-cli.md:809 and quickstart.md:992 all still state it is “**not a submodule of this tree**” — false as of this feature’s own work, never withdrawn (the constitution requires an explicit withdrawal, not a silent replacement).

Worse, the two are architecturally incompatible: pipeline-cli.md §4.8 specifies **native** launch (“run the built binary natively, as ai_interviewing does”), while T015 specifies **container** wrappers over cmd/boot.

Third strand: plan.md:96 records that the containers submodule’s own carriers were a §11.4.157(B) violation “which currently blocks the pre-push hook”, and no task clears it. (*Resolved 2026-09-01 upstream at 4dab992; the plan text is now stale in the other direction.*)

Resolution: one operator decision (native vs containerised), then withdraw the stale claim in all three documents and align pipeline-cli.md §4.8 with T015 or the reverse.

C2 — FR-039 (redaction) is a MUST with zero tasks

spec.md:219 requires a documented redaction step that MUST have run before any export or publication. It exists because the Chapter 1 recording features an identifiable third party.

It is **fully contracted**: pipeline-cli.md §4.6 (redact.sh) and §4.7 B5 (publish precondition); passage-contract.md §4.3 (redactions table), §7.3 rules R1–R7, gate G-PID-5; http-api.md §3.8 (410 Gone), §3.9 (redacted_omitted), §3.10 A4.

tasks.md has no redaction task. “redact” appears once (T064) as a *fixture kind*. Consequentially T071 describes add-chapter as “transcribe → ingest → index → cross-link” against pipeline-cli.md:559’s nine stages — dropping **preflight, verify-accuracy and redaction-review**, the last of which gates publication.

C3 — SC-016a has no implementing and no validating task

spec.md:258 requires code-passage identity keyed on symbol path with a rename alias table, such that 100% of stale code references **fail loudly**. Contracted at passage-contract.md §6.2–6.5, §4.3 (symbol_aliases DDL), §8 rule R4, gate G-PID-4.

T011 proves only G-PID-1/G-PID-2 (the two SC-016 survival guarantees). symbol_aliases, R4 and G-PID-4 appear in no task. SC-016a’s only mention is T081 — which *documents* a guarantee nothing implements.

Compounded by passage-contract.md:631 (P-U1): “Lumen exposes no symbol table through its CLI or MCP surface ... the matching key has no confirmed producer” — in no task and no register.

C4 — The Answer contract is forked

	data-model.md:144	http-api.md:824
field	verdict	status
refusal member	refused	declined
reason field	refusal_reason	reason
reason members	below_threshold, margin_too_small, unsupported, no_provider	below_threshold, margin_too_small, unsupported, no_citations, redacted_evidence

Four incompatibilities — and data-model.md files no_provider (an *unavailable* cause) as a **refusal** reason, conflating exactly the two states http-api.md:847 A7 declares distinct. plan.md:247 marks refusal and citation the two highest-risk [TDD] components, so tests land first against one of two shapes. T057–T068 cite http-api.md and never data-model.md.

HIGH (17, abbreviated)

#	Finding
H1	plan.md contradicts itself three times: Phase 0 “IN PROGRESS”/ Phase 1 “NOT STARTED” vs a post-Phase-1 re-check and 91 tasks; carriers “present and committed” vs “absent today”; [REVIEW] Container definitions vs “NO containers/ dir — deliberately NOT created”.
H2	SC-002 says “random sample of ≥ 30 passages”; research D-TRANS-3 samples audio-timeline windows (to remove deletion bias) — a reinterpretation repeated in four documents while tasks.md:26 claims values are “copied verbatim from spec.md”. “random” vs “seeded, stratified” are also different designs. The criterion was never amended.
H3	FR-016 requires indexing “the recordings themselves”; Lumen’s allowlist is a compile-time var with no override. Satisfied by <i>reduction to text</i>, which is a re-reading, not compliance. No task indexes a recording.
H4	passage-contract.md §6.6 quotes the pre-split SC-016 as “unqualified” and recommends a change already applied. plan.md:244 shares the stale reading and never mentions SC-016a; data-model.md mentions neither SC-016a nor symbol_aliases.
H5	Decision ids collide: research.md D-LLM-1...5 and llm-bridging.md

#	Finding
	D-LLM-1...11 mean different things (research D-LLM-1 = provider seam; llm-bridging D-LLM-1 = no generative model exists). T069 cites D-LLM-10 in the llm-bridging sense — valid but unresolvable from research.md, which tasks.md:27 names as its source. D-SEARCH-*/D-TRANS-* exist only in research.md.
H6	Control plane specified at two paths with two script sets. pipeline-cli.md:125 names six scripts and justifies via ai_interviewing's scripts/ — but the real reference is ai_interviewing/platform/scripts/. T015 creates 3 of 6; missing are restart.sh, build.sh and verify.sh — the aggregation point that runs every gate and implements SC-012/SC-013 precedence. Seven of eight contracted pipeline wrappers are untasked.
H7	T053 builds the SC-007/SC-008 benchmark; no task runs it. Compare T064/T065, which do it correctly. quickstart.md:620 specifies the missing bench-retrieval.sh. Both criteria are also BLOCKED: U5, stated flatly in Global Constraints with no caveat.
H8	SC-006 feasibility rests on an unverified assumption. tasks.md:29-31 juxtaposes “embedding 18.2–21.0 s” and “results ≤ 2 s p95” without reconciliation. search-architecture.md:377 asserts the

#	Finding
H9	2.2 s warm-up is “not paid per call” while its own C-3 marks that UNVERIFIED with counter-evidence (<code>index_status</code> cost 2.6–5.1 s on three consecutive calls in one warm session). No task settles U4/C-3, and the “reserved embedding capacity — a hard requirement” is untasked. Phase 3’s checkpoint cannot be met by Phase 3’s tasks: (a) the blind human reference (1–2 h) is contractually required and no task produces it, so T031 terminates at exit 2 by contract; (b) T030 claims to settle U1, but U1 is settled by the <code>venv</code> install — T005, thirteen tasks earlier — while T018–T029 presume a working faster-whisper with no failure branch; (c) the <code>whisper.cpp</code> fallback is itself unverified (U9, absent from <code>research.md</code>).
H10	The project constitution asserts two facts this feature measured false: “<code>ffmpeg/ffprobe</code> 7.0.2” (a Playwright symlink that fails <code>-show_format</code>), and “8 submodules are declared” (measured: 9) — against its own principle that “a hardcoded list silently goes stale the moment the set grows”. Root <code>CLAUDE.md</code>’s owned-submodule list omits both <code>workshop</code> and <code>containers</code>.
H11	<code>quickstart.md</code> states two counts of its own preflight failures — “Three of its six checks” vs “anything other than four 1s and two 0s has not understood the host”. The table shows four.

#	Finding
H12	FR-006 (preserve source material unmodified) has no task, though it is contracted with gate G-CLI-2. <code>plan.md:158</code> asserts the property as already true about a pipeline that does not exist.
H13	<code>sections[]</code> is served by the API (<code>http-api.md:333</code> , traced to FR-011) but modelled nowhere and built by no task. FR-011 is effectively untasked.
H14	The unverified register loses 17 of 22 entries and collides on two identifiers : transcription U4/U5 mean different things from research U4/U5. Transcription U6–U9 reach nothing downstream; two orphan UNVERIFIEDs sit in no register. Phase-1 contracts add P-U1/P-U2/P-U3 and C-U6/C-U7, all untasked — P-U2 in particular is the mechanism SC-016 rests on.
H15	T070 states the generation-latency estimates as fact, dropping the UNVERIFIED marker every other artifact carries — while the same task correctly forbids the API from publishing an estimate. The source contradicts itself: “engineering estimates, not measurements” vs “measured in tens of seconds”.
H16	SC-003’s “within 5 seconds” carries no statistic; downstream invents p95 in three places. Its feasibility rests on <code>transcription.md:724</code> ’s “ probably sufficient” — a hedge the governing rules forbid for

#	Finding
H17	causal claims — and U5 registers it unverified. SC-005's sole evidence (FTS5 p95 9.58 ms) was measured over the existing monorepo's 58,726 symbols, 92.7% of which are markdown headings — not curriculum passages, which do not exist. The architectural conclusion is robust; the <i>headroom</i> claim is not. Constitution: "A claim measured on one member of a set is NOT a claim about the set."

MEDIUM (22, abbreviated)

M1 paired-mutation inventory counted three ways (14 / "of 24" / actual 28), and **no task cites any G-* id** · M2 evidence path forked workshop/ evidence/ vs workshop/_evidence/ · M3 pipeline-cli.md:796 claims "42 FRs / 18 SCs" fully covered — there are 19 SCs and the claim is false for SC-015 and SC-016a · M4 "five corrected premises" does not decompose (premise 2 is "superseded by premise 5") · M5 "Open items entering Phase 1" lists three, one already resolved and still marked pending, and omits U3–U5 · M6 T055 proves SC-016 over citations built in a later phase · M7 checklist maps FR-039 to link-survival criteria · M8 provider env var forked, and the fallback makes the mismatch **silent** · M9 quickstart corrects a plan.md statement it no longer makes · M10 six contracted endpoints untasked · M11 data-model.md is a strict subset of the same-day contracts with no note that it is an abstraction · M12 SC-008's denominator ambiguous · M13 SC-004's 15-min budget has no host baseline while reassembly may dominate it · M14 SC-011's "hands-on time" undefined · M15 US3 called independently testable but declared to need US2 · M16 SC-018 has a writing task but nothing asserting 100% · M17 FR-029's progress half untasked · M18 FR-005's "state where it cannot" half untasked; speaker_source missing from the data model · M19 T017 inherits a two-valued script and adds no classifier, so an unreadable part reports 1 instead of 2 · M20 tombstones have no resolution outcome · M21 the withdrawn "75 s"

figure is restated as fact in `llm-bridging.md:657` · M22 `llm-bridging.md` defines two disjoint L1–L4 sets (grounding and privacy), so a bare “L3” is ambiguous.

LOW (6)

L1 `spec.md:21` “single word ‘Tbd.’” — measured: a heading plus Tbd. ·
L2 `workshop/docs/` marked NEW but exists; T001 must be non-destructive · L3 `WORKSHOP_CHUNK_SECONDS` is a flat 300 among derived neighbours · L4 registry path globbed two ways · L5 `pipeline-cli.md:702`’s example total matches no enumeration · L6 decision ordinal 2b round-trips through no `\d+` scheme.

Categories genuinely clean — reported as such

- **\$11.4.156 / no CI: clean.** Guarded at four independent points, including a gate whose mutation is “add one — must FAIL”. The workshop tree contains no `.github/`.
- **Every contracted gate carries a paired mutation: clean.** All 28 name a specific mutation and require it to redden. The real-entry-point rule and the grep-is-not-a-test rule are both carried through, and a gate whose mutation fails to redden it is defined as vacuous → FAIL. M1 concerns *counting* the inventory, not any gate lacking a proof.
- **Three-valued reporting: clean by design**, including a mechanical ERR trap mapping unclassified failures to 2. Only defect is M19, a task omission.
- **No frozen host assumptions in the design: clean.** Only L3 remains.
- **No scope creep: clean.** All 91 tasks trace to a requirement, decision, assumption or anchor.

Verified negatives (checked and sound): the 91-task count and per-phase split; T069’s D-LLM-10 is **not** dangling; `plan.md`’s “14 proofs” is an accurate count of quickstart’s table; the constitution’s “8 registered pre-push gates” (E + 0–6); the 36-part count.

Readiness verdict

NOT READY. Two MUST-level obligations (FR-039 redaction, SC-016a code identity) are fully contracted and entirely absent from 91 tasks — building now yields a system that cannot redact and that silently re-points code citations, the exact failures those requirements exist to prevent. Container orchestration is split across two incompatible architectures resting on a claim the tree disproves. The Answer contract is forked in the two components the plan itself marks highest-risk.

Suggested order: (1) the container decision — it changes T003, T015, `pipeline-cli.md` §4.8 and H6; (2) coverage gaps C2, C3, H12, H13, M10, M16–M19 — mechanical additions against contracts that already specify the work; (3) contract/data-model reconciliation C4, H4, M11 — must land before Phase 2's [TDD] work; (4) spec amendments H2, H3, H16, M12–M14, M7; (5) open-item handling H7–H9, H14 — consolidate the register, add settling tasks, assign the human reference; (6) consistency and count corrections.

Every CRITICAL and HIGH finding is a document reconciliation or a missing task — none requires rethinking the architecture. Two need more than editing: C1 needs an operator decision, and H9(a) needs the 1–2 hours of blind human transcription nobody has been assigned.

A note on why this analysis was possible

The artifacts are unusually strong where it matters most: the three-valued instrument design, the measured rejection of Lumen chunk ids and the ULID scheme built in its place, the D-TRANS-3 deletion-bias argument, the egress-denied negative control, and above all the unverified registers. The documents repeatedly refuse to claim what they have not measured — **including about themselves:** `plan.md` withdraws its own `ffprobe` claim, `spec.md:145` withdraws the 75 s figure. That discipline is the only reason this analysis could find drift at all. In a set of documents that bluffed, none of these 49 contradictions would have been visible.